

Business Requirements Document (BRD)

Next-Generation Autonomous Voice AI Agent Platform & White-Label Infrastructure

Project:	QuickVoice Enterprise Platform
Document Version:	v2.5.0 (Production Release)
Status:	Approved for Implementation
Target Audience:	Engineering, Product, DevOps, Agency Partners
Date:	September 2026

1. Executive Summary & Product Vision

1.1 Business Context & Problem Statement

Modern enterprise contact centers and B2B client operations face critical bottlenecks:

- **Excessive Operational Costs:** Human agent labor costs average \$1.20 - \$2.50 per operational minute with high agent attrition.
- **Rigid Legacy IVR Systems:** Traditional "press 1 for sales" phone trees result in customer frustration, abandoned calls, and negative brand perception.
- **Latency Disconnect in First-Gen Voice Bots:** Traditional multi-agent pipelines introduce 1,500ms - 2,500ms response delays, resulting in unnatural conversational pauses.
- **Lack of Multi-Tenant Reseller Capabilities:** Agencies and enterprise resellers cannot customize, re-brand, or white-label existing voice AI platforms without extensive engineering overhauls.

1.2 The QuickVoice Solution

QuickVoice is an end-to-end, sub-600ms latency autonomous Voice AI platform built on WebRTC and real-time generative models. It combines dynamic speech-to-speech interaction, deterministic state machines, live supervisor coaching, warm SIP/PSTN transfers, and complete white-label multi-tenancy for digital agencies and enterprises.

Core Value Proposition

QuickVoice empowers companies to deploy human-like voice agents capable of conducting appointment scheduling, inbound support intake, outbound sales qualification, and automated CRM updates at 80% lower cost than traditional call centers.

1.3 Strategic Objectives & Success Metrics (KPIs)

Metric	Baseline (Legacy/Competitors)	QuickVoice Target	Business Impact
End-to-End Voice Latency	1,200ms - 2,200ms	< 650ms	Imperceptible conversational pause; natural conversational flow.
Cost per Call Minute	\$0.75 - \$1.80	\$0.06 - \$0.12	85%+ operational expenditure reduction.
First-Contact Resolution (FCR)	45%	78%+	Drastic reduction in escalations and repeated callbacks.
White-Label Onboarding Time	2-3 weeks (custom fork)	< 5 minutes (No-Code)	Immediate tenant deployment for agency reseller networks.
System Availability SLA	99.5%	99.95%	Carrier-grade reliability for enterprise voice workloads.

2. Stakeholders & User Personas

2.1 Stakeholder Matrix

Stakeholder Group	Key Interests & Priorities	Success Criteria
Agency Owners / Resellers	Full platform white-labeling, custom domains, branding isolation, client markup.	Zero QuickVoice branding visible; instant tenant onboarding; custom accent colors.
Call Center Supervisors	Call monitoring, live audio interception, real-time agent whisper coaching, escalation control.	Sub-second silent audio listen-in; seamless prompt coaching during live calls.
Voice Prompt Engineers	Visual flow state machines, slot collection, prompt compilation, deterministic guardrails.	Visual node builder synced directly to system prompt without raw prompt sprawl.
DevOps / Infrastructure Engineers	Dockerized microservices, low resource footprint on VPS, secure secrets management, automated DB sync.	Single <code>docker compose</code> deployment; automated migrations; health checks.
Compliance & Security Officers	Zero-PII retention, AES-256 encrypted credentials, GDPR/HIPAA compatibility, signed webhooks.	Configurable conversation retention days; zero audio recording retention mode.

2.2 Key User Personas

Persona 1: Sarah — Agency Founder

Goal: Provide AI receptionist services to 40 dental and law practices under her agency brand "ApexVoice".

Pain Point: Competitor platforms display their own logos and don't allow custom domains.

QuickVoice Benefit: Custom logo, custom color swatch, custom subdomain, and total watermark removal in 3 clicks.

Persona 2: Marcus — Call Center Lead

Goal: Monitor complex customer inquiries and guide the AI when unexpected questions arise.

Pain Point: Inability to hear what the AI is saying in real-time or intervene before customer hangs up.

QuickVoice Benefit: Live WebRTC Listen-In and Whisper Coaching console to guide the agent mid-sentence.

Persona 3: Elena — Head of Customer Support

Goal: Ensure VIP callers or angry customers are transferred to tier-2 human specialists seamlessly.

Pain Point: Cold transfers drop calls or force the caller to repeat their issue from scratch.

QuickVoice Benefit: Warm SIP REFER transfer with pre-transfer speech, fallback messages, and forwarded summaries.

Persona 4: David — Lead Integrations Architect

Goal: Sync caller information into GoHighLevel, HubSpot, and trigger instant confirmation SMS.

Pain Point: Building custom webhooks and schemas for every client tool takes weeks.

QuickVoice Benefit: 1-Click integration templates with pre-configured schemas and automatic prompt bindings.

3. System Scope & Technical Architecture

3.1 High-Level Architecture Overview

The QuickVoice platform consists of three decoupled operational tiers:

- 1. **Presentation Tier (Console):** Next.js 16 (React 19, Turbopack, Tailwind CSS 4) web application providing agent management, visual state machines, real-time call docks, analytics, and white-label settings.
- 2. **Application & Orchestration Tier (Server):** Node.js / Express API server managing Better Auth, organization access controls, telephony webhook routing, BullMQ background queues, and PostgreSQL data persistence.
- 3. **Real-Time Media & Voice Engine (AI Worker):** Python service interfacing with LiveKit WebRTC server, Silero Voice Activity Detection (VAD), Deepgram STT, LLM inference models (Anthropic Claude, OpenAI, Groq), and Deepgram/ElevenLabs TTS.

3.2 System Architecture Diagram (Component Interaction)

Layer	Technology	Role & Responsibility
Frontend UI	Next.js 16, TanStack Query, Radix UI, Lucide	Dynamic White-Label Tenant Console, Flow Builder, Live Coaching Dock.
API Gateway / Server	Express 5, Node.js 20+, Prisma 7, Socket.IO	REST Endpoints, Swagger Docs, Live Transcript Gateway, Webhook dispatching.
Data Persistence	PostgreSQL 16 & Redis 7	Relational entities, audit logs, financial ledger, BullMQ job queues, Pub/Sub channels.
Real-Time Audio Plane	LiveKit (WebRTC / SIP)	Ultra-low latency audio streaming, SIP trunking to Twilio/Telnyx carrier networks.
Voice AI Pipeline	Python 3.11, Deepgram Nova-3, Claude/GPT-4o, Deepgram Aura	Streaming Speech-to-Text, LLM turn generation, streaming Text-to-Speech synthesis.
Storage & Security	AWS S3 / Cloudflare R2, AES-256	Encrypted call audio recordings, Knowledge Base documents, token encryption.

3.3 Media vs. Signaling Plane Separation

Architectural Advantage

Control signaling (call setup, prompt updates, webhook notifications, and database writes) travels over lightweight HTTP and WebSockets. Voice audio packets never traverse the API server; they route directly over WebRTC UDP through LiveKit to the Python voice worker, guaranteeing sub-600ms latency.

4. Detailed Functional Requirements Specification

4.1 Feature Breakdown (FR-01 through FR-07)

FR-01: Multi-Tenant White-Labeling & Reseller Branding Engine

- **Priority:** Must Have (P0)
- **Requirement:** The platform must enable organization owners to fully re-brand the console interface without modifying code or restarting containers.
- **Inputs:** App Name, Logo URL, Favicon URL, Primary Hex Color, Custom Host/Domain, Powered-By Toggle.
- **Outputs:** Dynamic `:root` CSS variables (`--primary` , `--ring`), browser page title, favicon metadata, and customized logo components.
- **Acceptance Criteria:**
 - Tenant settings saved under `Organization.metadata.branding`.
 - Hex color is dynamically converted to HSL color space to match theme contrast standards.
 - Disabling "Powered By QuickVoice" hides all vendor watermarks and footer badges across console and web widget.

FR-02: Visual Conversation Flow & State Machine Builder

- **Priority:** Must Have (P0)
- **Requirement:** Provide a node-based visual drag-and-drop state machine builder for structuring deterministic voice agent dialogue trees.
- **Node Types:**
 1. `start` : Initial greeting and conversational entry point.
 2. `message` : Informational speech turn without slot collection.
 3. `question` : Slot collection turn with strict data type validation (String, Phone, Date, Email).
 4. `condition` : Multi-branch decision logic based on slot values or caller intent.
 5. `action` : Automated API tool execution (e.g., booking appointment, checking database).
 6. `transfer` : Escalation to external human phone number or SIP trunk.
 7. `end` : Wrap-up speech and polite call termination.
- **Prompt Compiler ("Sync to Prompt"):** Converts the entire node graph into structured system prompt instructions injected directly into the agent configuration.

FR-03: Human-in-the-Loop Warm Call Transfer (SIP REFER & PSTN)

- **Priority:** Must Have (P0)
- **Requirement:** Allow voice agents to transfer live telephone calls to human representatives over PSTN (E.164) or SIP trunks with zero audio drop.

- **Features:**

- Configurable pre-transfer announcement message (e.g. "Please hold while I transfer you to our specialist.").
- Fallback handling if destination line is busy, unanswered, or fails.
- Caller trigger phrases (e.g., "speak to human", "manager", "representative").
- Context forwarding: Call summary and caller sentiment are attached to the SIP payload/webhook.

FR-04: Real-Time Silent Audio Listen-In & Whisper Coaching

- **Priority:** High (P1)

- **Requirement:** Supervisors must be able to silently monitor live active calls and whisper guidance to the agent in real time.

- **Capabilities:**

- **Listen In:** Subscribes to the agent's WebRTC audio track with microphone muted, allowing real-time listening.
- **Whisper to Agent:** Sends high-priority text instructions to the agent's LLM context via LiveKit data channel; the agent incorporates the instruction into its immediate next speech turn without the caller knowing.

FR-05: 1-Click Ecosystem Integration Templates

- **Priority:** High (P1)
- **Requirement:** Provide ready-to-use integration templates that eliminate manual JSON schema configuration for popular business tools.
- **Pre-Built Integrations:**

Integration	Category	Action Executed by Voice Agent
Cal.com	Scheduling	Checks calendar availability and books meeting slots live during call.
GoHighLevel (GHL)	CRM & Sales	Creates/updates contact records, assigns pipeline tags, and logs call outcome.
HubSpot CRM	Enterprise CRM	Enriches caller properties (name, email, phone) and creates qualified sales deals.
Zendesk Support	Helpdesk	Generates incident ticket containing call summary, transcript, and priority.
Twilio SMS Follow-up	Messaging	Dispatches instant SMS booking links or confirmation texts to caller's mobile.

FR-06: Automated Agent Evaluator & Benchmark Simulator Studio

- **Priority:** High (P1)
- **Requirement:** Enable developers to benchmark and test voice agents against synthetic caller personas prior to deploying live telephony numbers.
- **Built-In Personas:**
 - *Alex (Appointment Client):* Cooperative, straightforward, provides required slots promptly.
 - *Jordan (Angry Customer):* Frustrated, demands refunds, tests guardrails and escalation logic.
 - *Taylor (Vague Lead):* Indecisive, asks ambiguous questions, tests clarification handling.
 - *Eve (Edge-Case Adversary):* Tries to jailbreak the agent, off-topic prompts, tests safety guardrails.
- **Scorecard Metrics:** Goal Completion Rate (%), Conversational Latency (ms), Sentiment Score, Guardrail Integrity (Pass/Fail).

FR-07: Financial Ledger & Micro-USD Wallet Billing

- **Priority:** Must Have (P0)
- **Requirement:** Support prepaid wallet balances metered at micro-USD (\$0.000001) precision with automatic reservations and reconciliation.
- **Specifications:**
 - Prevents negative balances: Outbound/inbound calls check wallet reservation before connecting.
 - Stripe Wallet Checkout for automated credit top-ups.
 - Detailed per-second itemized transaction ledger (telephony, STT, LLM, TTS breakdown).

5. Non-Functional Requirements (NFR)

5.1 Performance & Latency Budgets

Pipeline Stage	Maximum Latency Budget	Optimizations Applied
Voice Activity Detection (VAD)	< 50ms	Silero VAD running locally on audio frame buffers.
Speech-to-Text (STT)	< 180ms	Deepgram Nova-3 streaming WebSocket endpoint with interim results.
LLM First Token Generation	< 200ms	Claude 3.5 Haiku / Groq Llama 3 / GPT-4o-mini streaming token output.
Text-to-Speech (TTS) First Chunk	< 150ms	Deepgram Aura / Cartesia streaming audio synthesis on first token sentence chunk.
Total End-to-End Latency	< 580ms - 650ms	Pipelined chunk-based streaming execution across all stages.

5.2 Security & Data Privacy

- **Zero-PII Retention Option:** Toggling `zero_pii_retention: true` automatically strips caller transcript recordings, names, and phone numbers immediately after call completion.
- **Configurable Retention Days:** Organizations can set automated call transcript and audio pruning policies (e.g. 7, 30, or 90 days).
- **Data Encryption:** All sensitive credentials (API keys, tool secrets, webhook signing keys) are encrypted at rest using AES-256-GCM via a dedicated 32-byte `SECRET_ENCRYPTION_KEY`.
- **Transport Security:** All external API calls and WebRTC streams mandate TLS 1.3 and SRTP (Secure Real-time Transport Protocol).

5.3 High Availability & Disaster Recovery

- **Stateless Voice Workers:** Python AI workers run as stateless agents. If an instance restarts, active calls can gracefully fall back to backup instances.
- **Queued Asynchronous Tasks:** Knowledge base ingestion and outbound dialer queues operate on Redis-backed BullMQ with automated exponential backoff retries.
- **Health Check Endpoints:** `GET /api/v1/system/readiness` monitors database, redis, and provider connectivity for container orchestrators.

6. Implementation Roadmap & Milestones

Milestone	Deliverables & Focus Areas	Status
Phase 1: Core Engine & Telephony	WebRTC pipeline, Deepgram STT/TTS, Twilio/Telnyx SIP trunks, LiveKit bridge, basic console.	✓ Complete
Phase 2: Enterprise White-Labeling	Multi-tenant branding provider, custom domains, logo/favicon dynamic injection, theme switcher.	✓ Complete
Phase 3: Visual Flow & Human Handoff	Visual state machine builder, prompt compiler, warm SIP/PSTN call transfer, live supervisor whisper.	✓ Complete
Phase 4: Integrations & Testing Studio	1-Click integration templates (Cal.com, GHL, HubSpot, Zendesk), automated evaluation studio.	✓ Complete
Phase 5: Self-Hosted Cloud Rollout	Production Docker Compose stack, VPS deployment guides, comprehensive BRD & Operator Manuals.	✓ Complete

7. Sign-Off & Approval

This Business Requirements Document has been reviewed and approved by all key stakeholders:

Role	Name	Signature / Approval	Date
Product Lead	QuickVoice Product Team	APPROVED (Digital)	September 2026
Lead Solutions Architect	Core Systems Architecture	APPROVED (Digital)	September 2026
Security & Compliance Lead	SecOps & Infrastructure	APPROVED (Digital)	September 2026